AGT1 - MODULE-2

Interactive visualization at: https://mehrsachal.github.io/InteractiveLearning/#/map-algebra-companion

Objectives

- Understand Map Algebra
- Use Raster Calculator tool
- Use Int tool
- Use Arithmetic operators
- Use Boolean And
- Use Combinatorial And
- Use Over tool
- Use Diff tool
- Use InList Tool

Deliverables (To be submitted)

- Map showing use of each tool/operator. (Screenshot **not** Map Layout)

Lab Settings

Use Data in folder: **AGT1-Data/Module2**

Scenario

Imagine you are on a mission to restore the lush forests of Waziristan.

To succeed, you have to play the role of a **"Nature Architect"** using the language of **Map Algebra** to decode the landscape.

Think of Map Algebra as a superpower that lets you stack different layers of the Earth like a mathematical sandwich to reveal secrets hidden in the terrain.

Raster Calculator

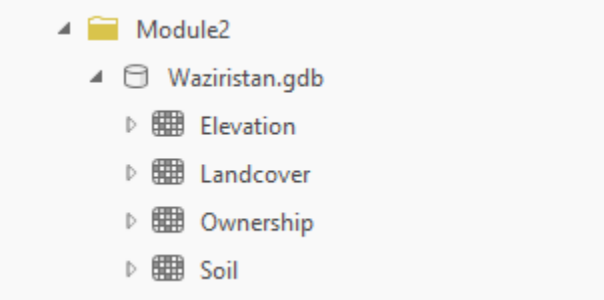
The **Raster Calculator** is a powerful spatial analysis tool in ArcGIS (available in both ArcMap and ArcGIS Pro) that allows you to perform complex mathematical calculations on raster datasets using **Map Algebra**.

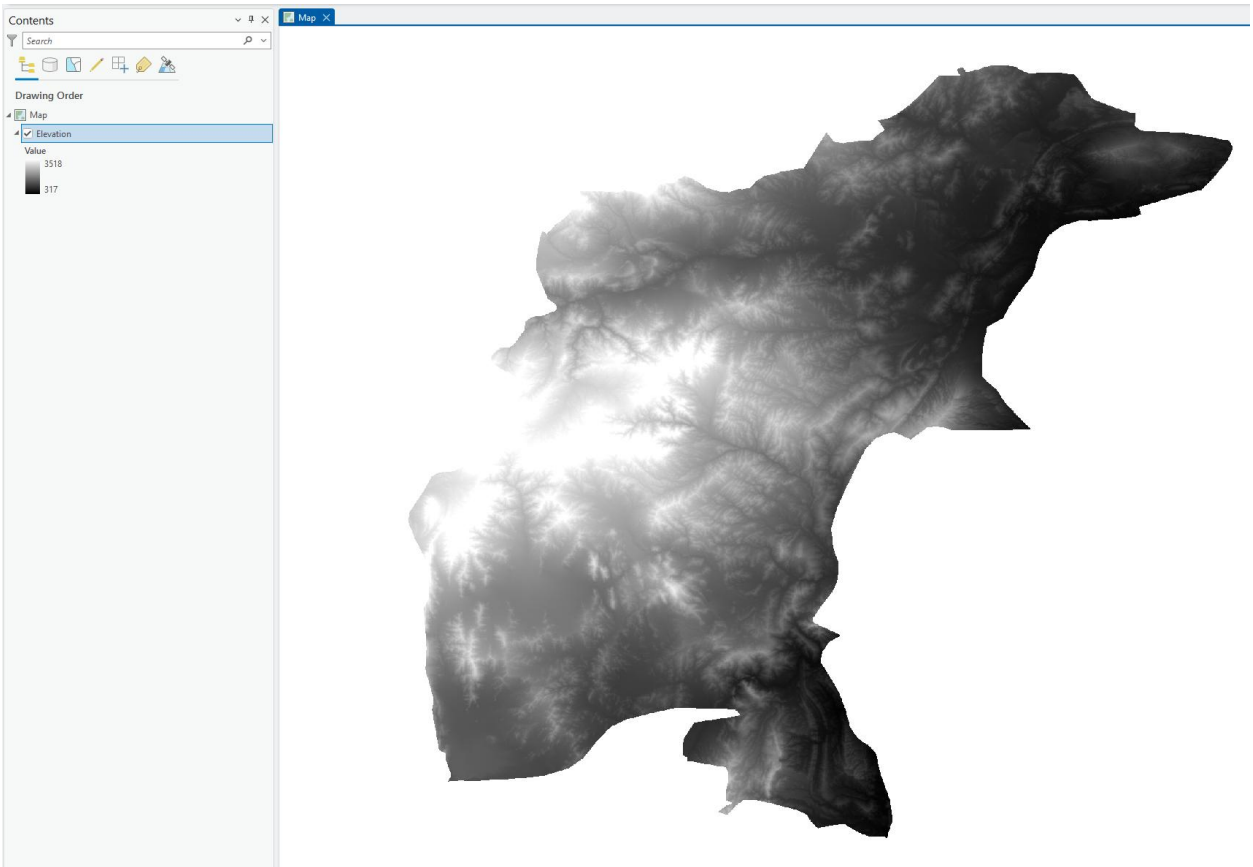
Think of it as a scientific calculator where the "numbers" you input are actually entire grids of geographic data.

Goal: Use Raster calculator.

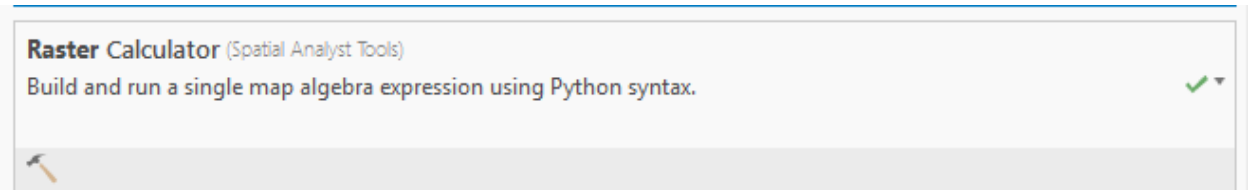
Scenario: The elevation data is initially recorded in feet, but the project requires metric units for volume and suitability calculations. You use the **Times** (multiply) operator to multiply the elevation by 0.3048.

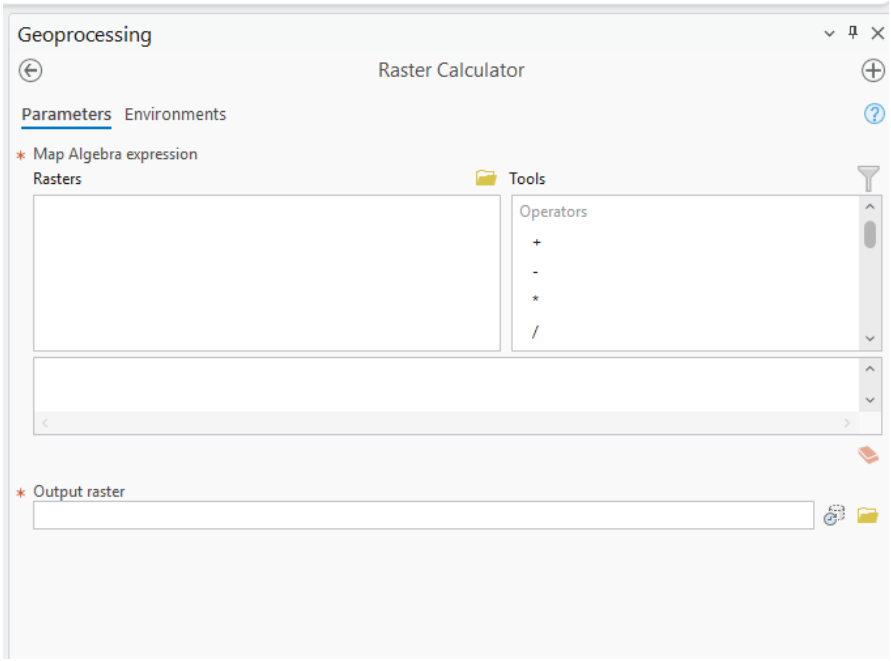
Add Elevation (Raster) Data:





Open **Raster calculator** (spatial analyst):

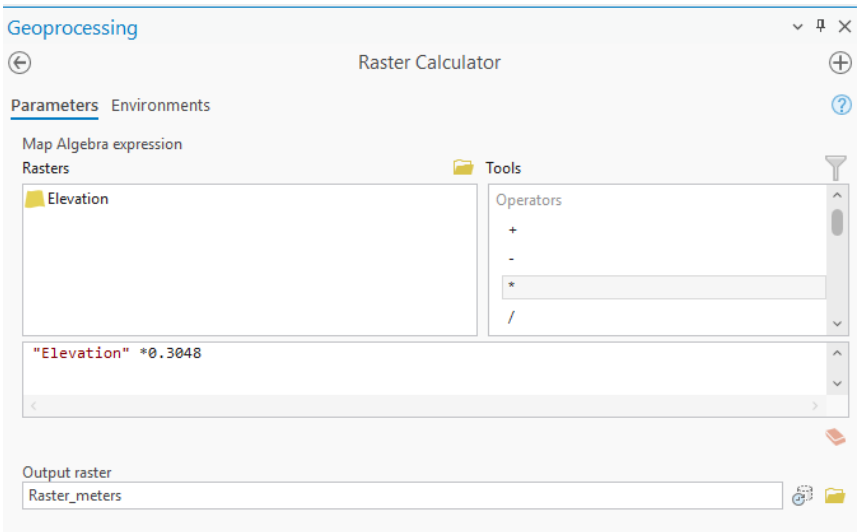




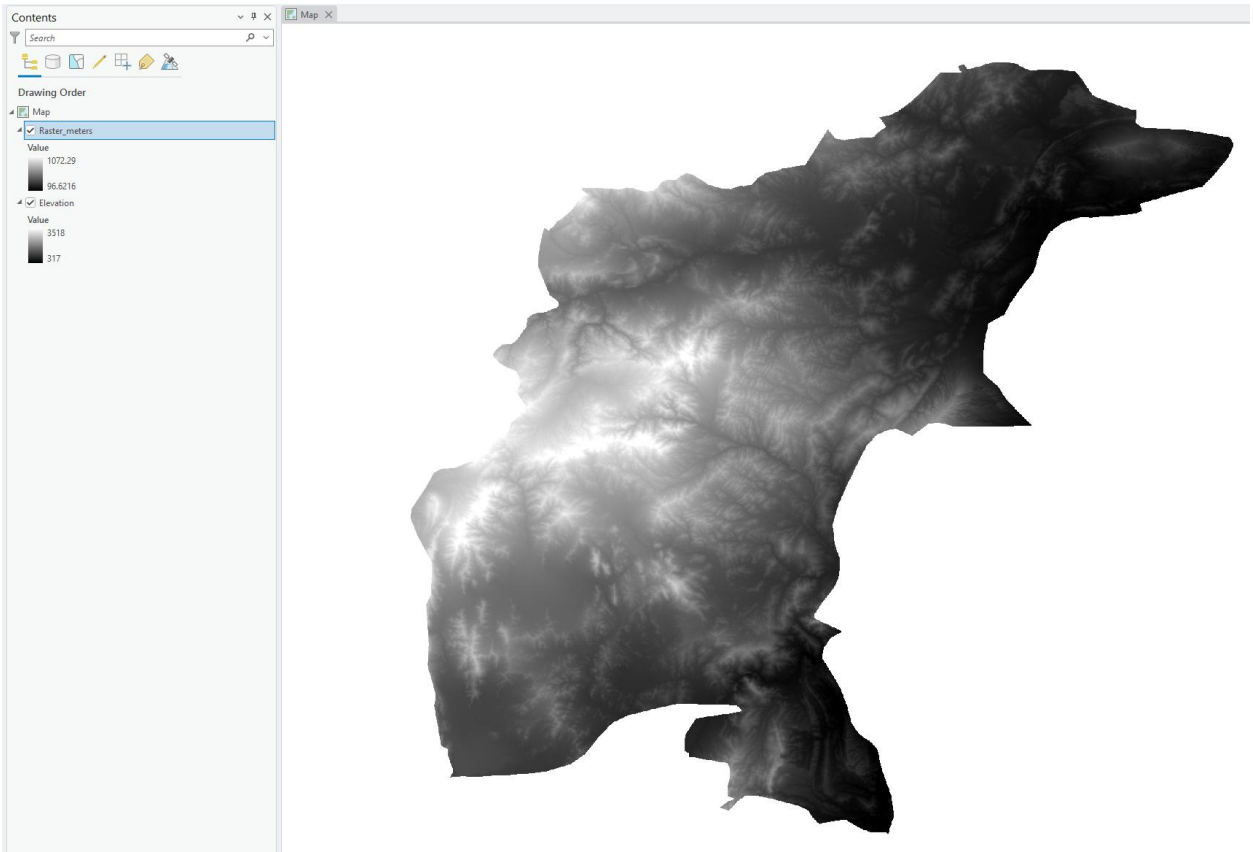
Use the variables and operators available to write the following expression:

"Elevation" *0.3048

Note: Double-clicking on a variable or operator adds it to the expression.



Hit **Run:**



A new DEM with values in meters has been generated. (Note the change in values ion the legend).

Int Tool

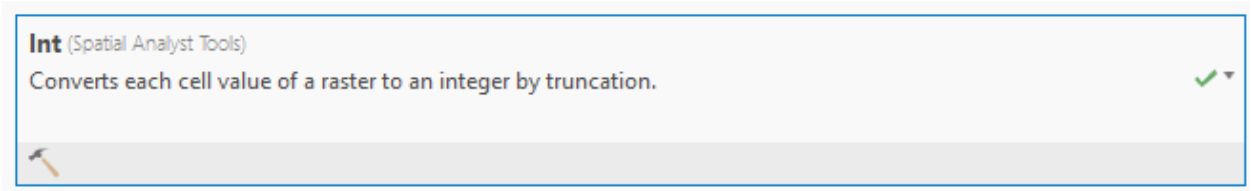
The **Int** tool converts a floating-point raster into an integer raster by **truncating** the decimal values (e.g., 5.9 becomes 5).

Its primary purpose is to transform continuous data into discrete categories, which allows ArcGIS to generate a **Raster Attribute Table** for further analysis. This is essential when you need to perform operations that require integer inputs, such as calculating the area of specific classes.

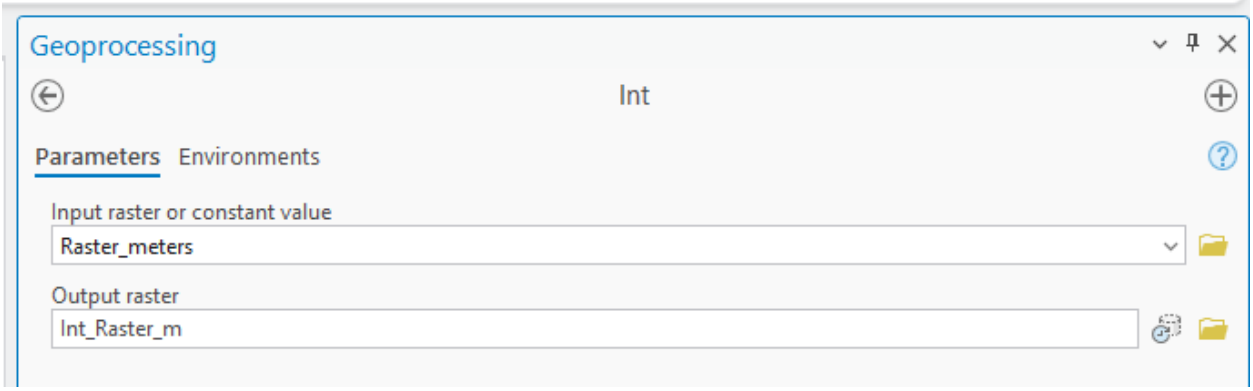
Goal: Use Int Tool.

Scenario: To keep the data clean for processing, the Int function is then used to truncate those results into whole integers.

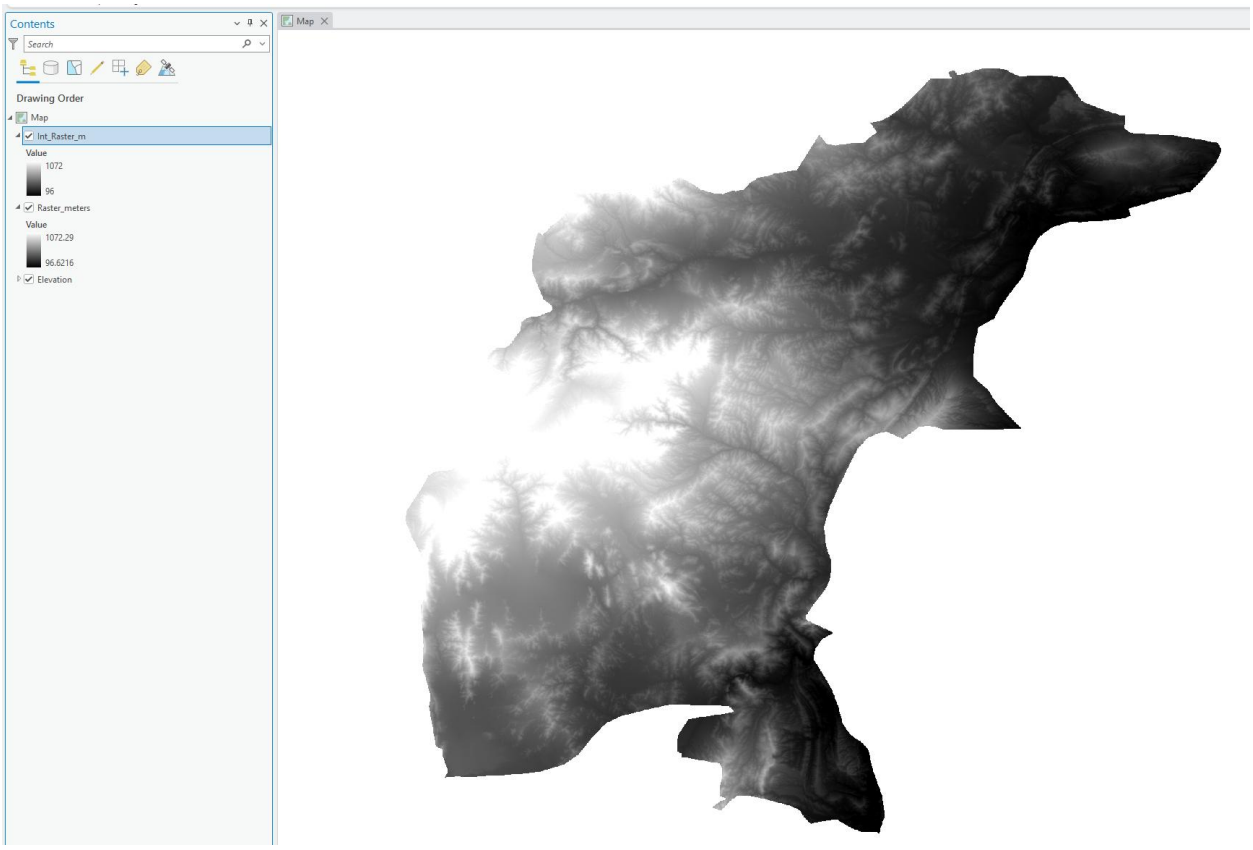
Open **Int** tool:



Configure it as follows:

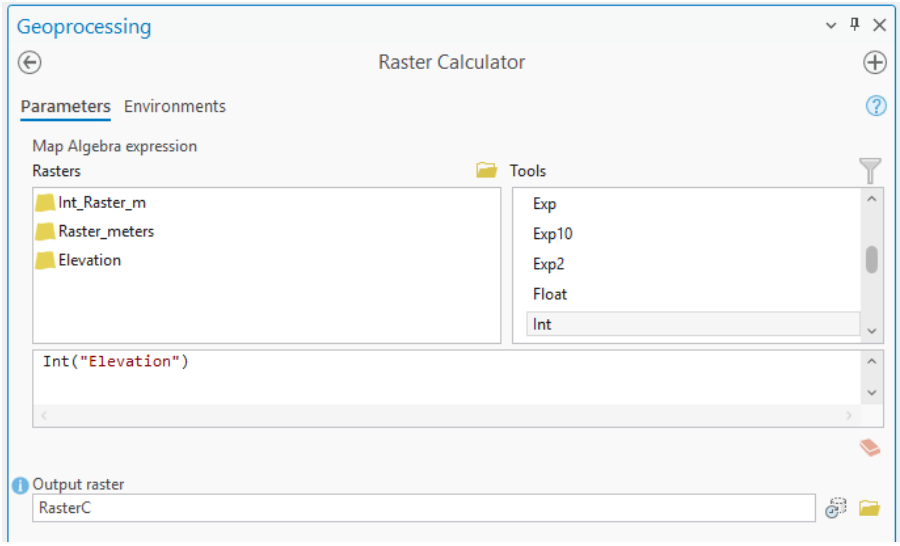


Hit **Run:**



Note that the values in Legend have been truncated (Rounded Off).

<u>Note:</u> We can perform the same in Raster calculator as well!



Greater Than Tool

The **Greater Than (>)** tool performs a logical comparison that identifies which cells in a raster are larger than another value or raster.

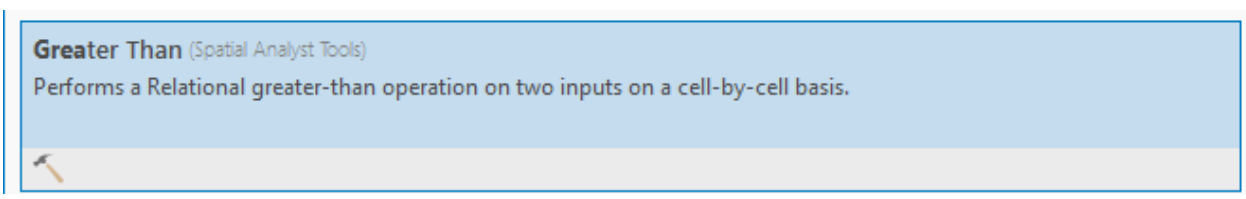
It outputs a binary result, where cells meeting the condition are assigned **1 (True)** and those that do not are assigned **0 (False)**.

This is a fundamental operation for masking data or isolating specific features, such as identifying terrain with an elevation higher than a set threshold.

Goal: Use **Greater than** Tool.

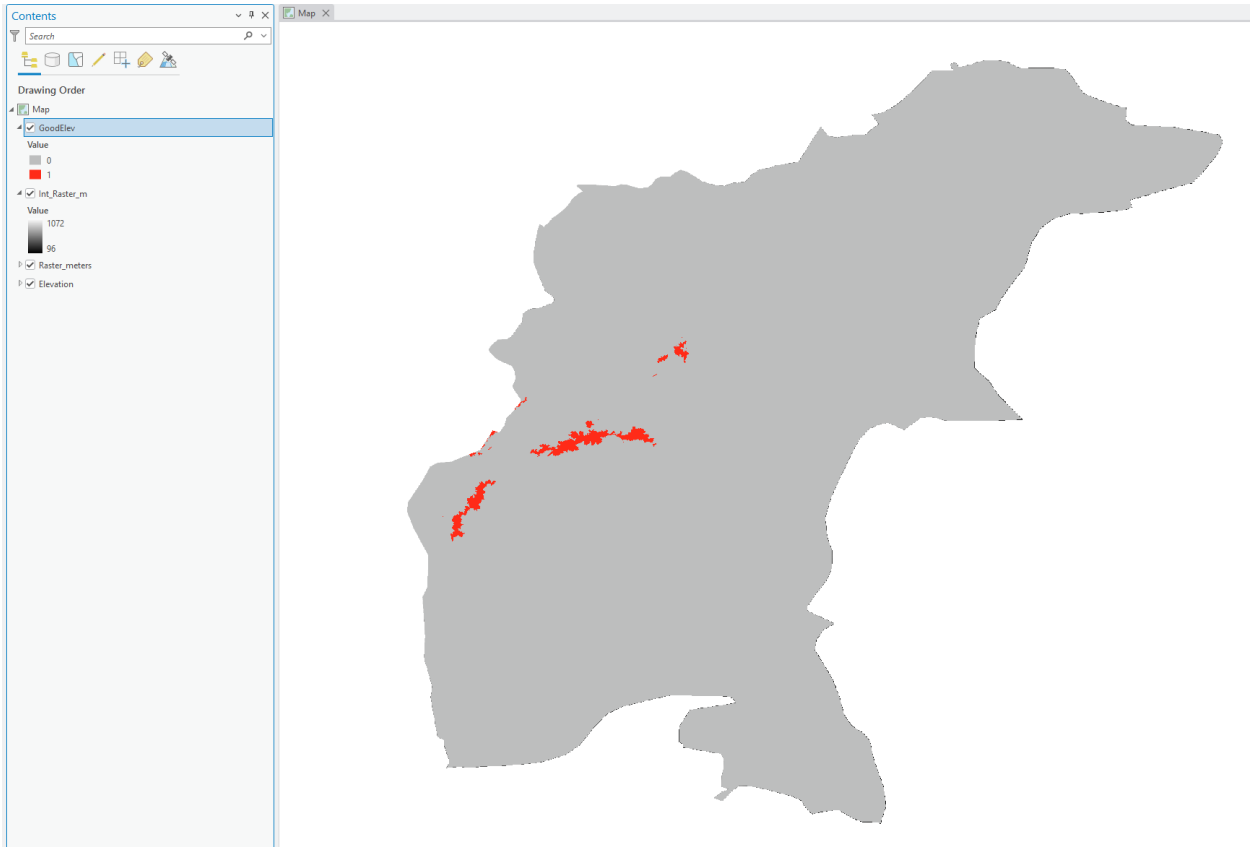
Scenario: The target pine and fir trees grow best at altitudes above **900 meters.**

Open **Greater Than** tool.

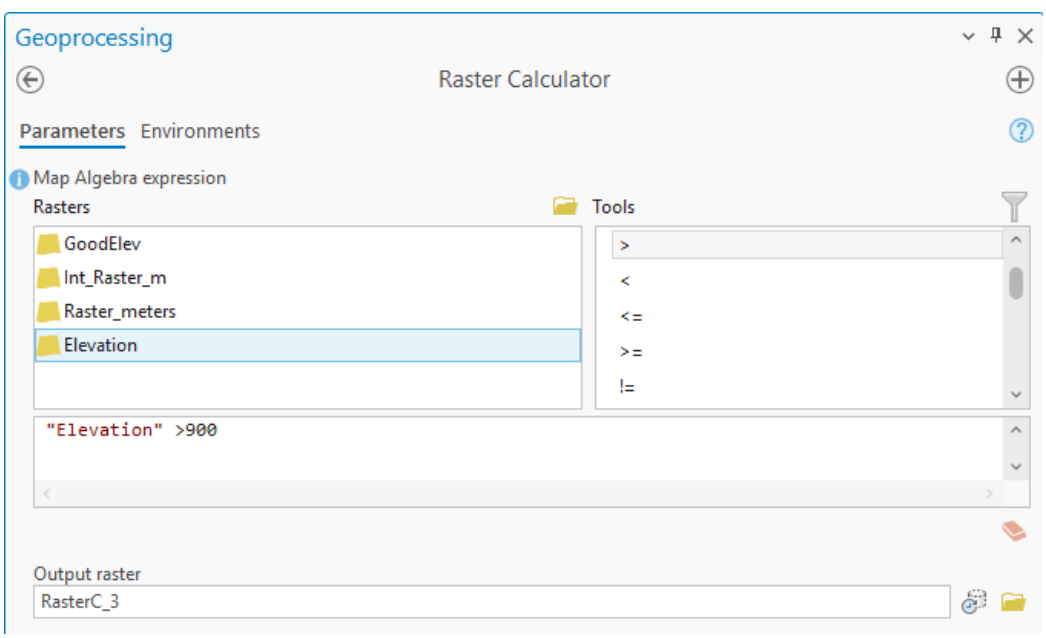


Configure as follows:

Hit **Run:**



<u>Note:</u> We can perform the same in Raster calculator as well!



Equal To Operator

The **Equal To (==)** tool performs a cell-by-cell comparison to determine if values in two rasters (or a raster and a constant) are identical.

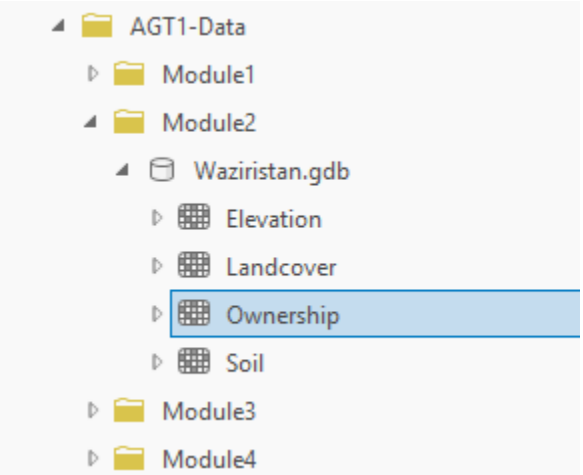
It produces a boolean output where matching cells are assigned **1 (True)** and non-matching cells are assigned **0 (False)**.

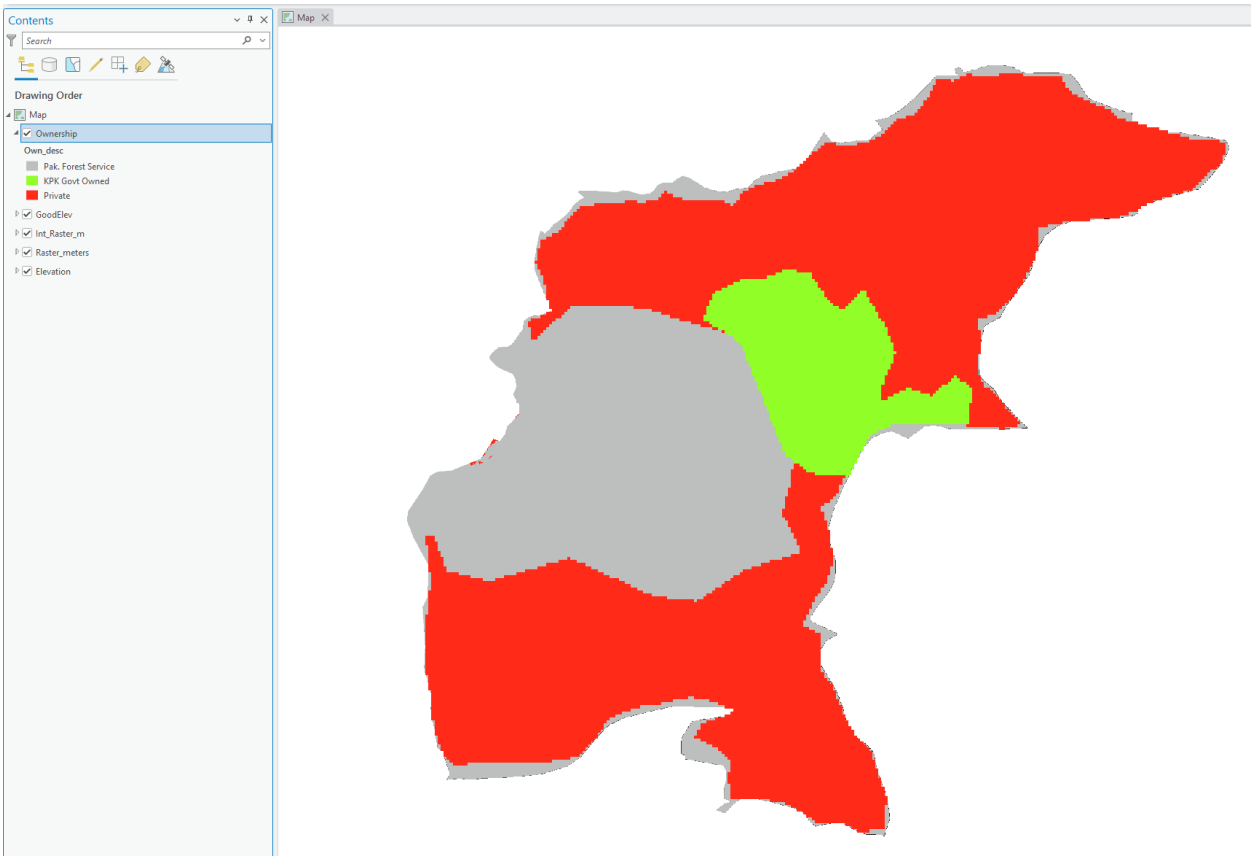
This operator is frequently used in land-use analysis to isolate specific categories or to create binary masks for spatial modeling.

Goal: Use **Equal To** Operator.

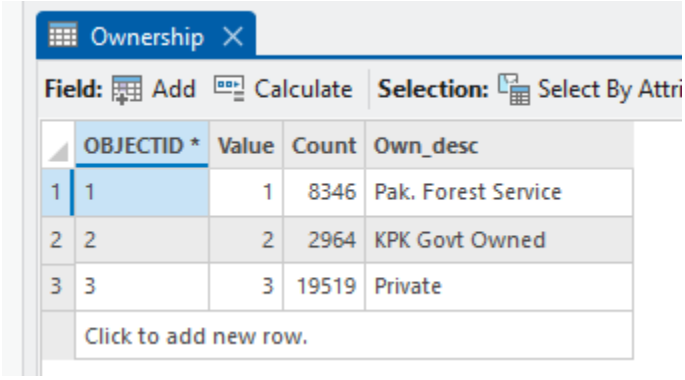
Scenario: You must also ensure the land is owned by the **Pakistan Forest Service.**

Load **Ownership (Raster):**





Explore the raster's attributes:



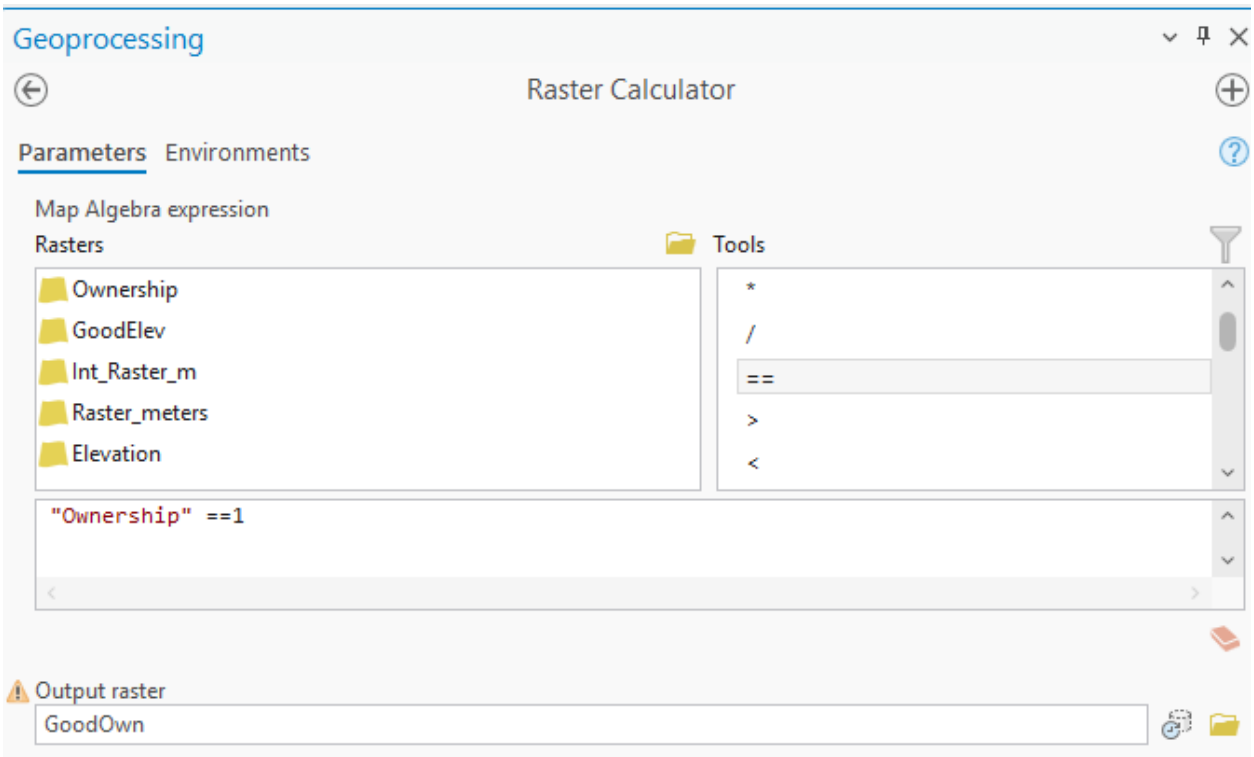
Ownership

Field: Add Calculate Selection: Select By Attri

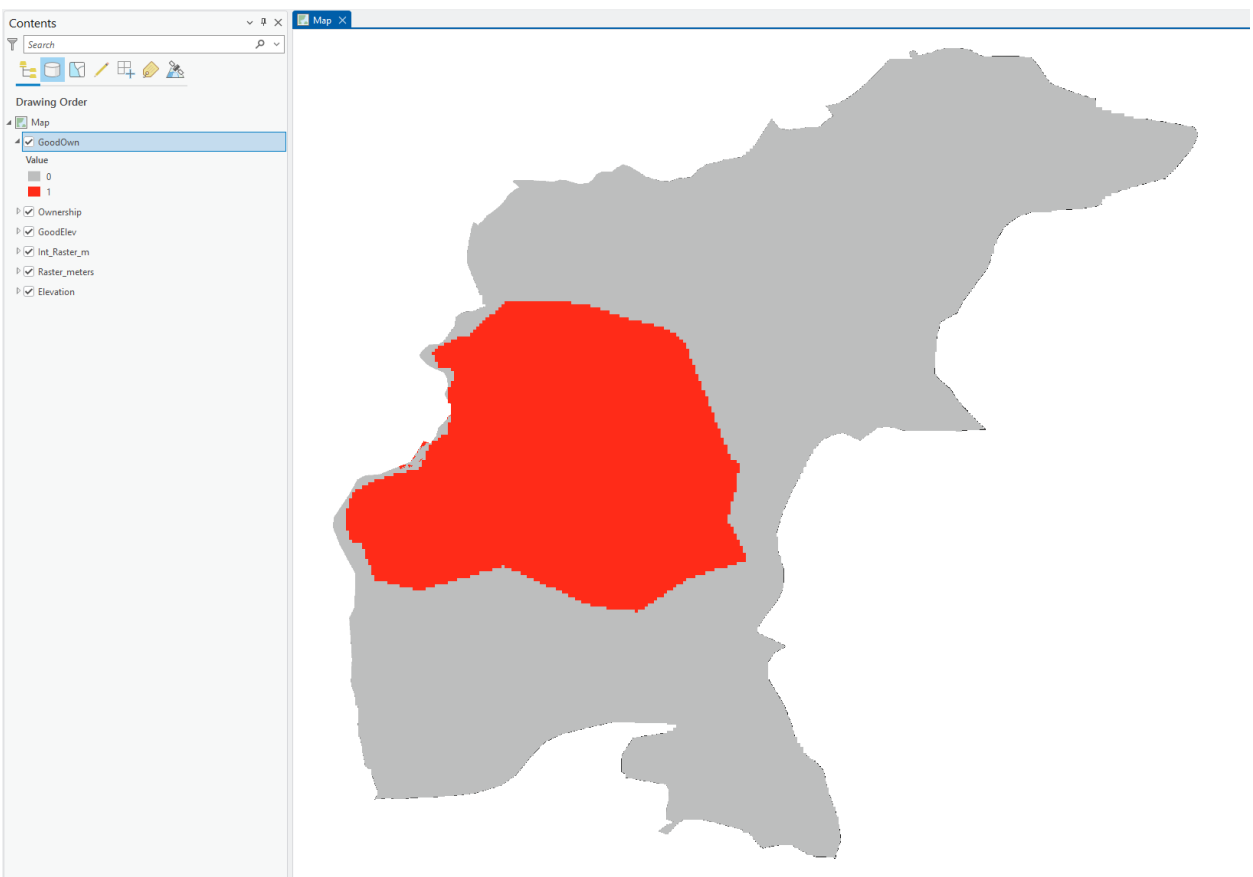
	OBJECTID *	Value	Count	Own_desc
1	1	1	8346	Pak. Forest Service
2	2	2	2964	KPK Govt Owned
3	3	3	19519	Private
	Click to add new row.			

We see that a value of 1 indicates that the land is owned by **Pak Forest Service.**

We will now use the **equal to** operator to identify such cells as **True** (1):


Geoprocessing
Raster Calculator
Parameters
Environments
Map Algebra expression
Rasters
Tools
Ownership
GoodElev
Int_Raster_m
Raster_meters
Elevation
*
/
==
>
<
"Ownership" ==1
Output raster
GoodOwn

Hit Run:

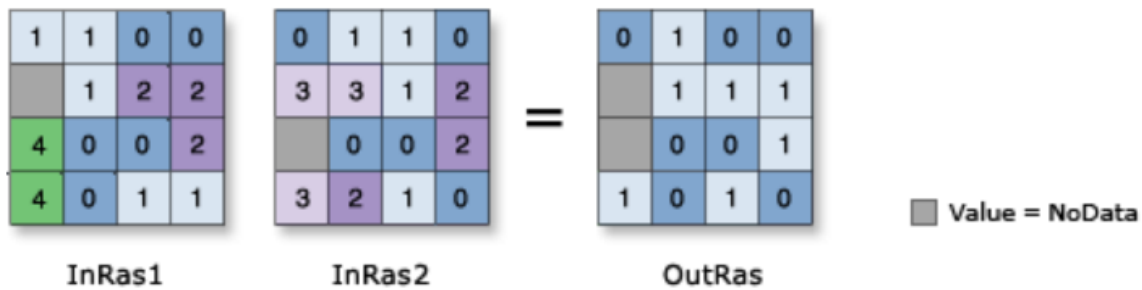

Contents
Search
Drawing Order
Map
GoodOwn
Value
0
1
Ownership
GoodElev
Int_Raster_m
Raster_meters
Elevation
Map

Boolean And Tool

The **Boolean And (&)** tool performs a logical evaluation where a cell in the output is assigned **1 (True)** only if the values in **both** input rasters are non-zero.

If either (or both) input cells are zero, the output is assigned **0 (False)**.

It is essentially an "intersection" operation, commonly used in suitability modeling to find areas where multiple required criteria are met simultaneously.

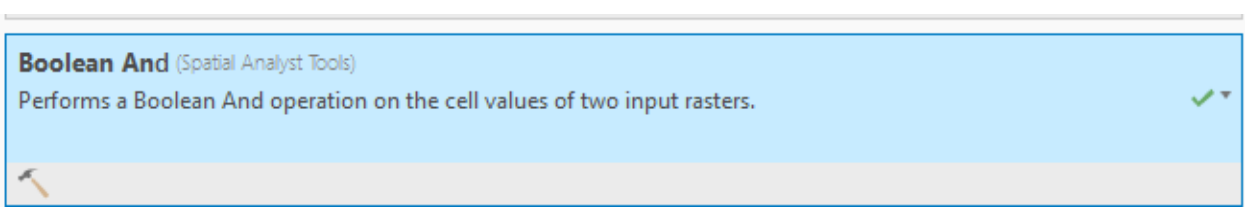



```
OutRas = BooleanAnd(InRas1, InRas2)
```

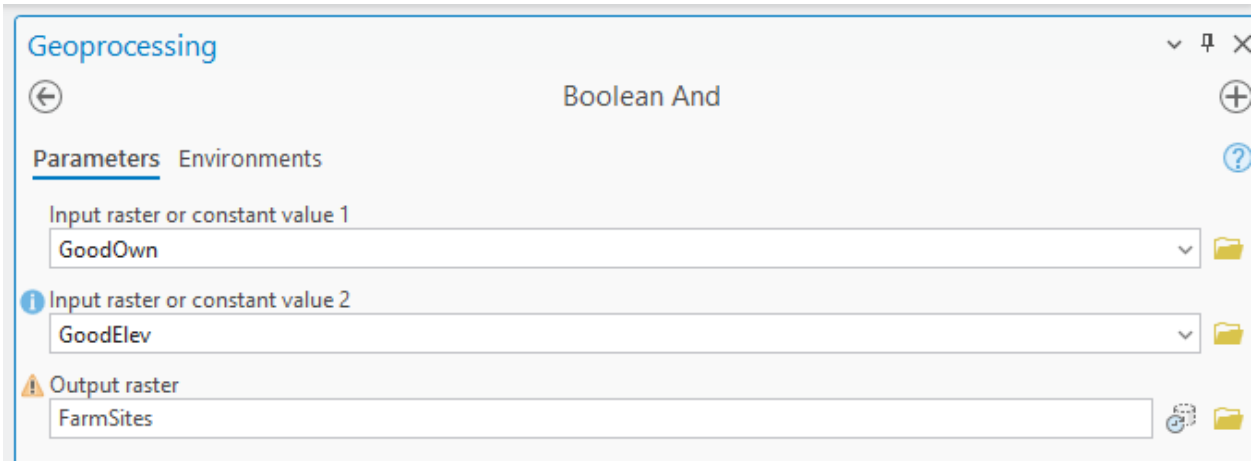
Goal: Use **Boolean And** Operator.

Scenario: Find areas where both requirements (height & ownership) are met **simultaneously.**

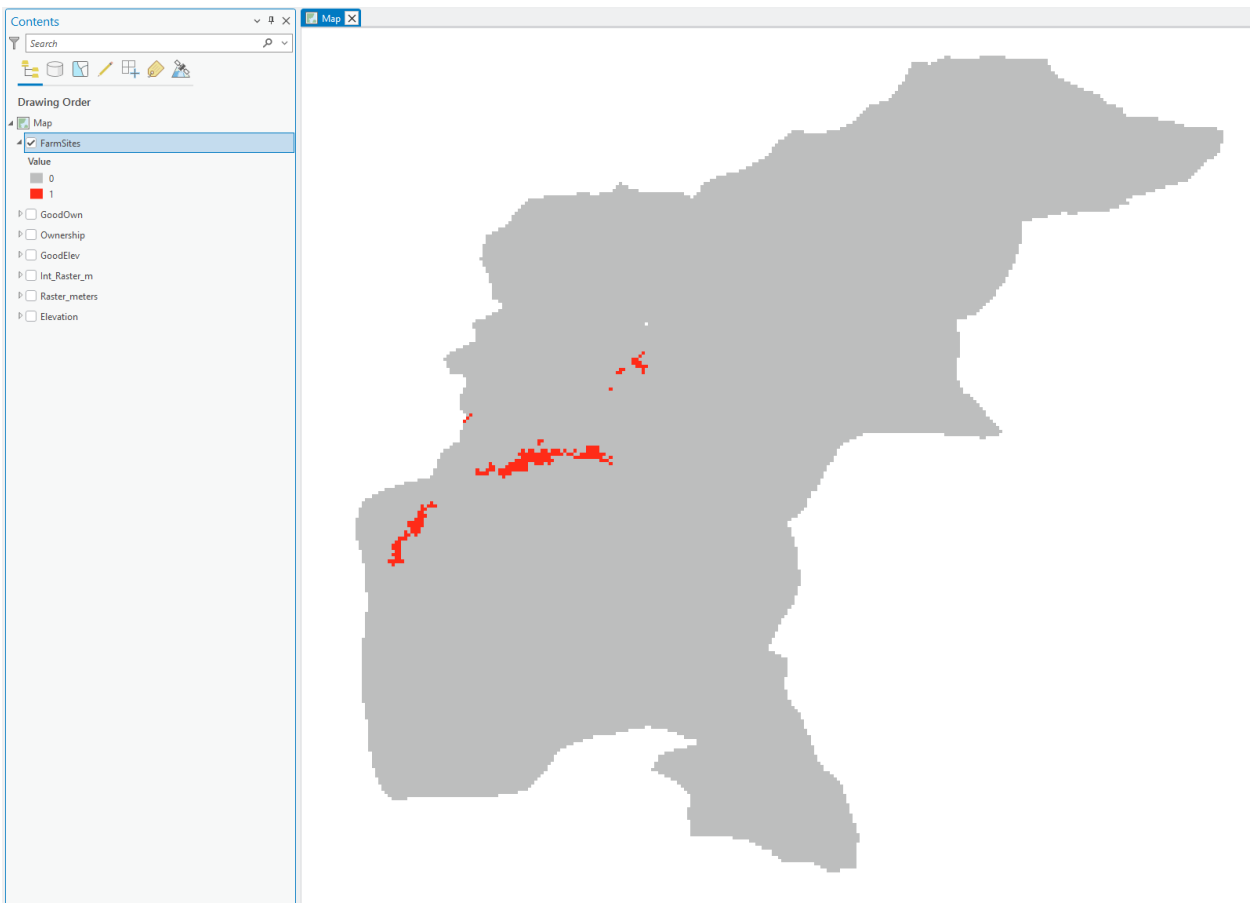
Open **Boolean AND** tool:



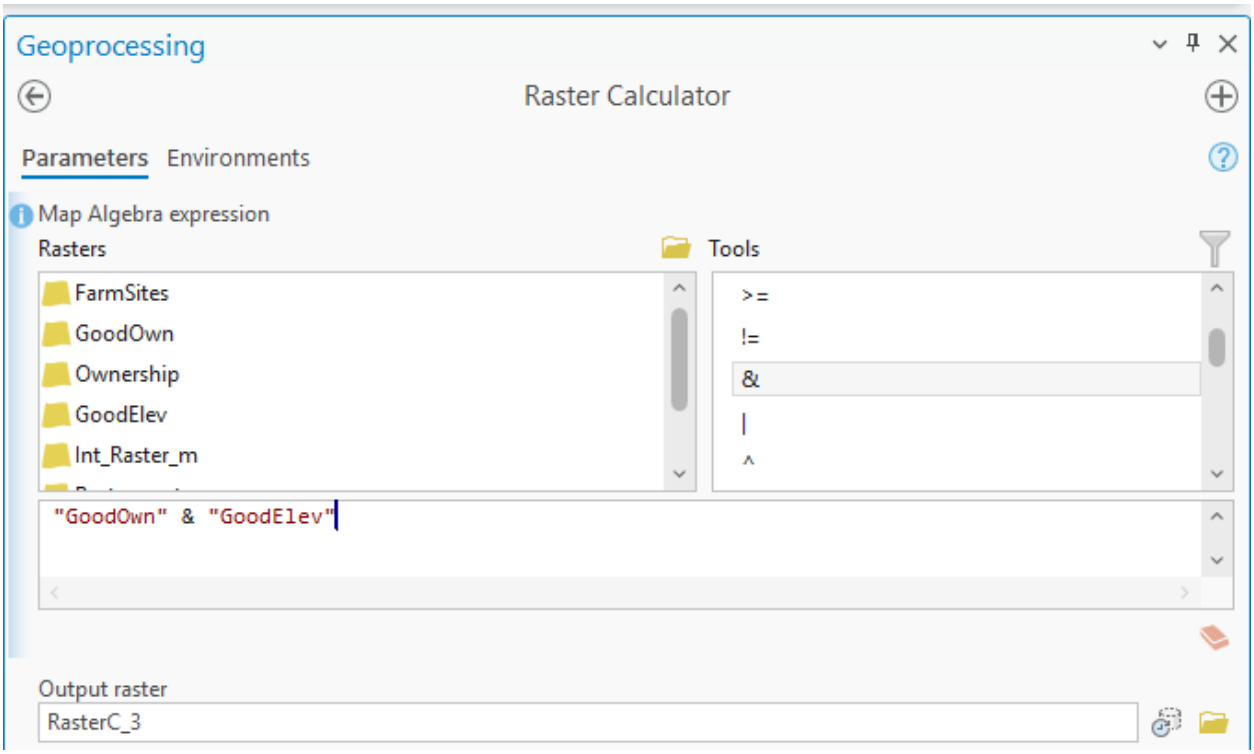
Configure as fol:



Hit **Run.**



<u>Note:</u> We can perform the same in Raster calculator as well!

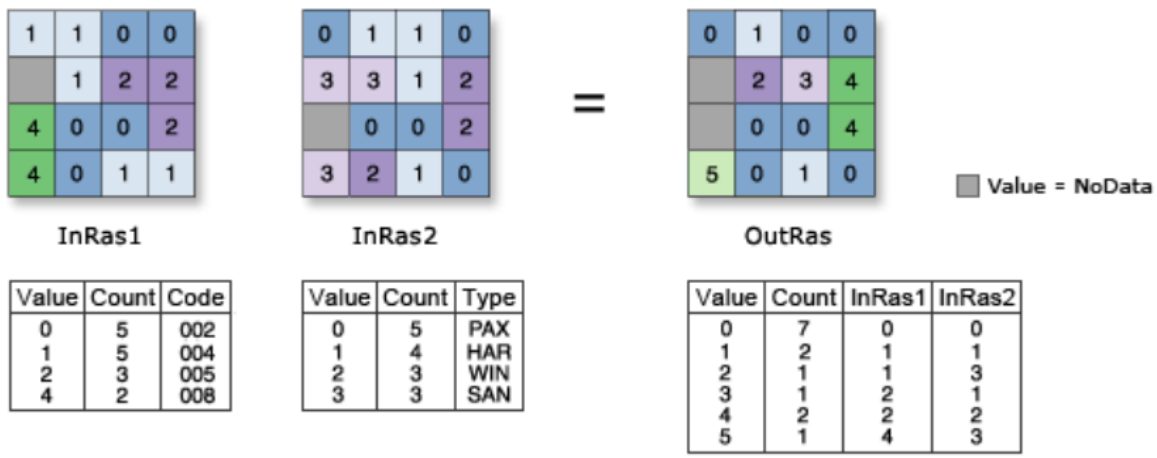


Combinatorial AND

The **Combinatorial And (CAND)** tool performs a cell-by-cell logical evaluation of two or more rasters, but unlike **Boolean AND**, it preserves the **unique combinations** of the input values.

While **Boolean AND** only tells you if both inputs are true (1) or false (0), CAND assigns a **unique integer** to **every** distinct overlapping pair of input values and records these combinations in an attribute table.

This allows you to track exactly which specific attributes from the original layers are present in each cell of the output, making it superior for complex multi-criteria overlays.



Value	Count	Code
0	5	002
1	5	004
2	3	005
4	2	008

Value	Count	Type
0	5	PAX
1	4	HAR
2	3	WIN
3	3	SAN

Value	Count	InRas1	InRas2
0	7	0	0
1	2	1	1
2	1	1	3
3	1	2	1
4	2	2	2
5	1	4	3

```
OutRas = CombinatorialAnd(InRas1, InRas2)
```

Imagine you have two different maps, and you want to find the spots where **both** maps have something special.

Boolean AND is like a simple "Yes/No" filter. It only cares if both maps have *anything* there. If Map A has a tree and Map B has a rock, the output just says "Yes, both have something." It throws away the details and only gives you a 1 (Yes) or a 0 (No).

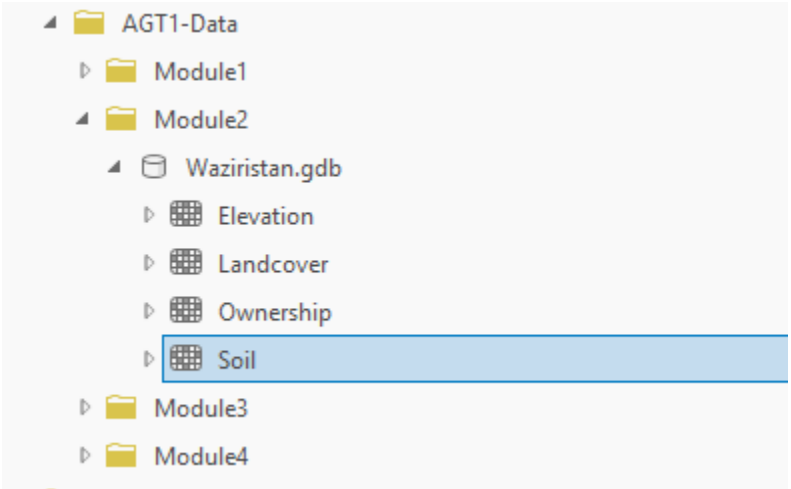
CAND (Combinatorial And) is like a "Memory" filter. It doesn't just say "Yes"; it remembers exactly what was on each map.

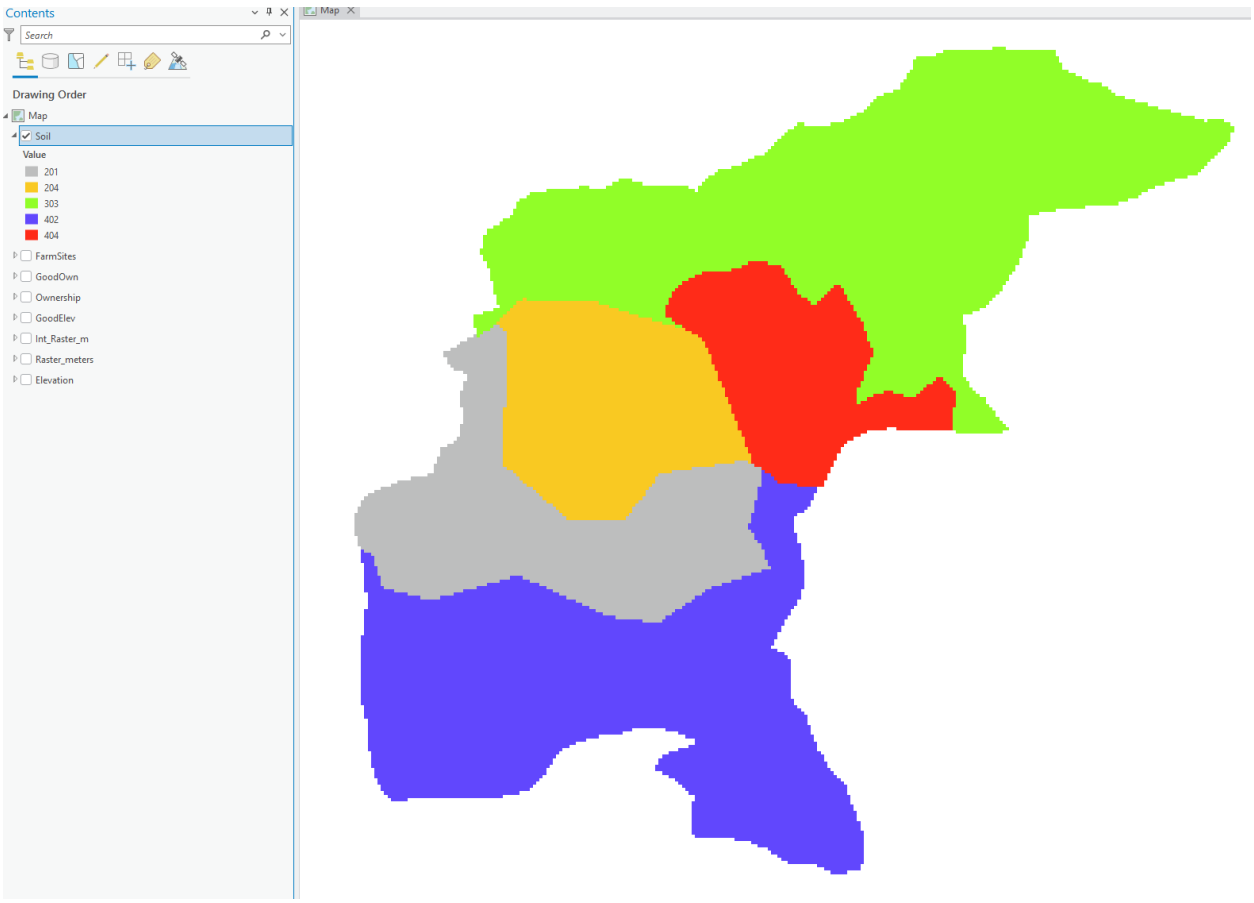
If Map A has a "Pine Tree" and Map B has a "Granite Rock," CAND creates a new category called "Pine Tree + Granite Rock." It keeps a list (an attribute table) of every unique combination it finds, so you don't lose the specific details from your original maps.

<u>Goal:</u> Use **CAND (Combinatorial AND) Tool**.

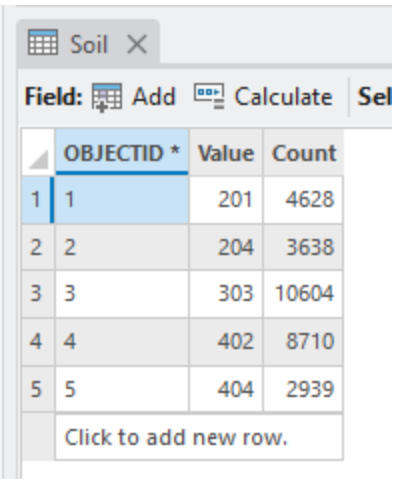
<u>Scenario:</u> Now that we have selected our Farm Sites, add soil attributes to them.

Load **Soils (Raster):**

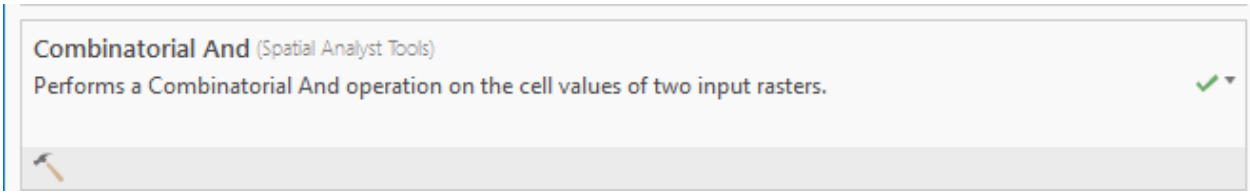




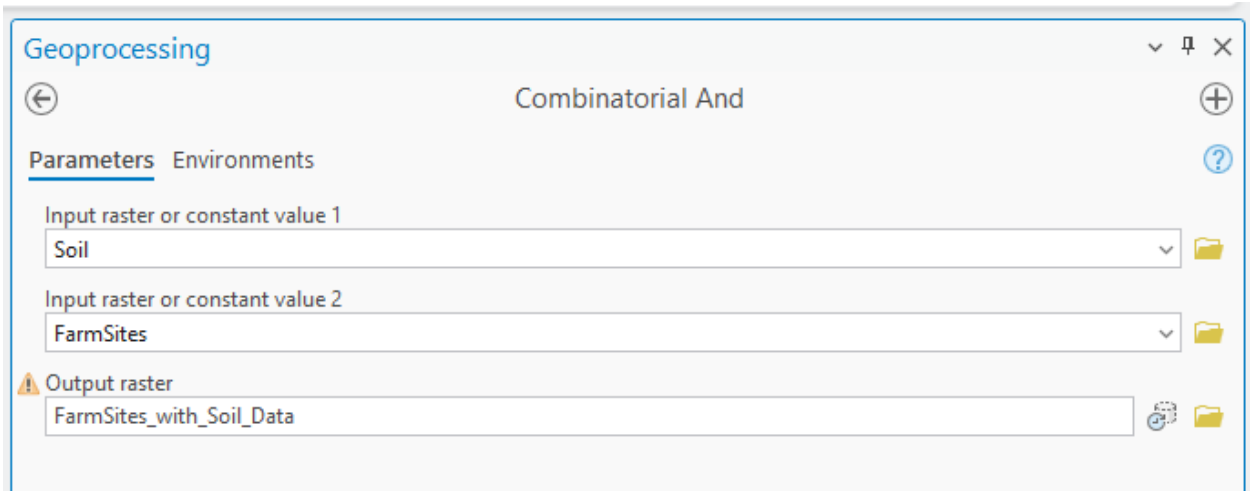
Explore attribute table:



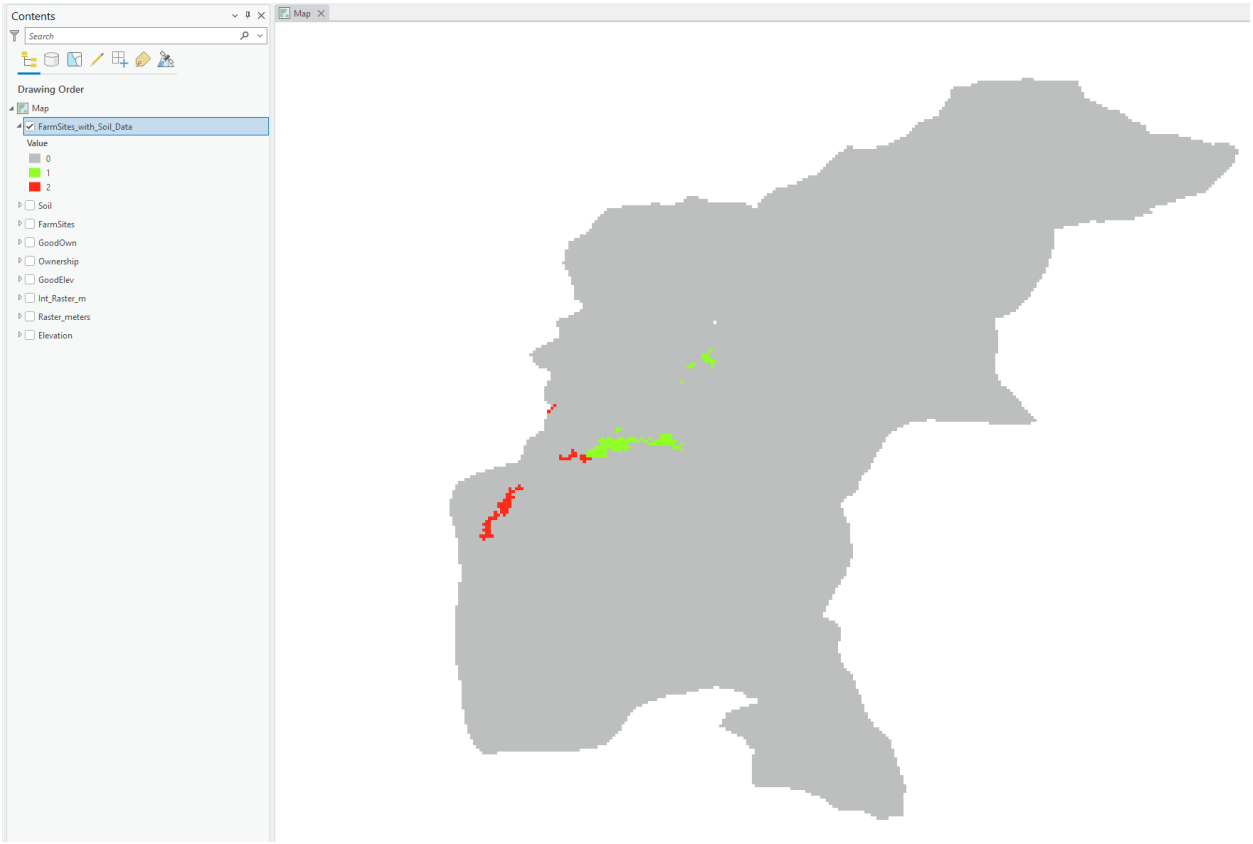
Open Combinatorial AND tool:



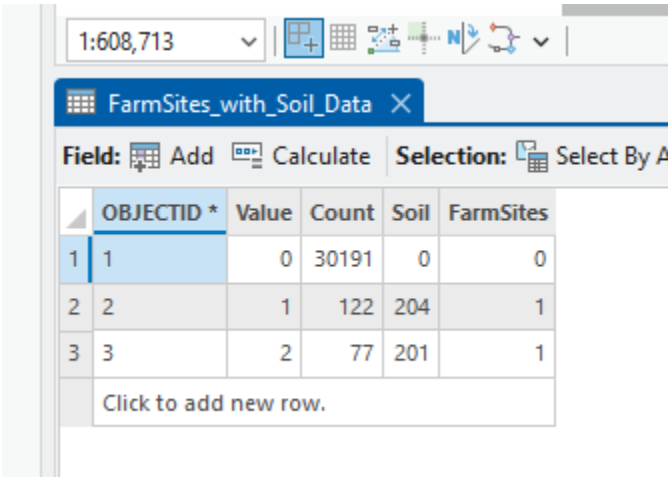
Configure as follows:



Hit **Run:**


Contents
Search
Drawing Order
Map
FarmSites_with_Soil_Data
Value
0
1
2
Soil
FarmSites
GoodOwn
Ownership
GoodElev
Int_Raster_m
Raster_meters
Elevation
Map

Let’s explore attributes:



	OBJECTID *	Value	Count	Soil	FarmSites
1	1	0	30191	0	0
2	2	1	122	204	1
3	3	2	77	201	1

Notice how a unique value was created for every valid combination.

Over Tool

The **Over** tool works like a physical **rubber stamp**, where the values of the first input raster are "stamped" on top of the second input raster.

If the first raster has valid data (non-zero or non-null values), those values are kept in the output; however, wherever the first raster has a 0 (zero), the tool allows the values from the second raster to "show through".

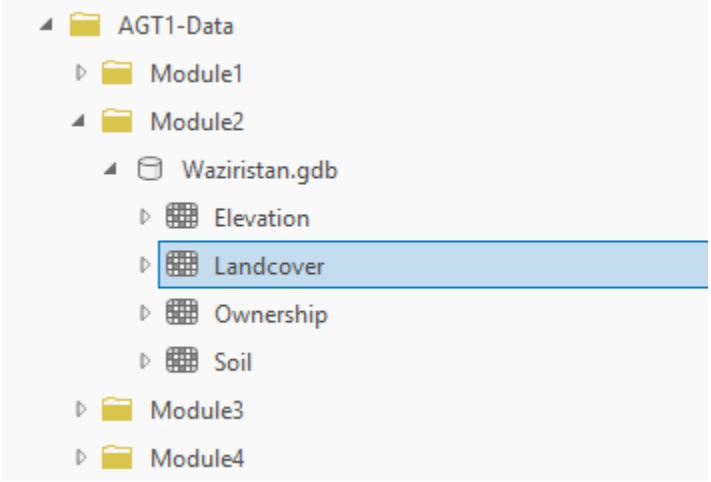
This makes it an ideal tool for patching holes in a dataset or layering a new feature over an existing background while keeping the underlying data visible wherever the top layer is empty.

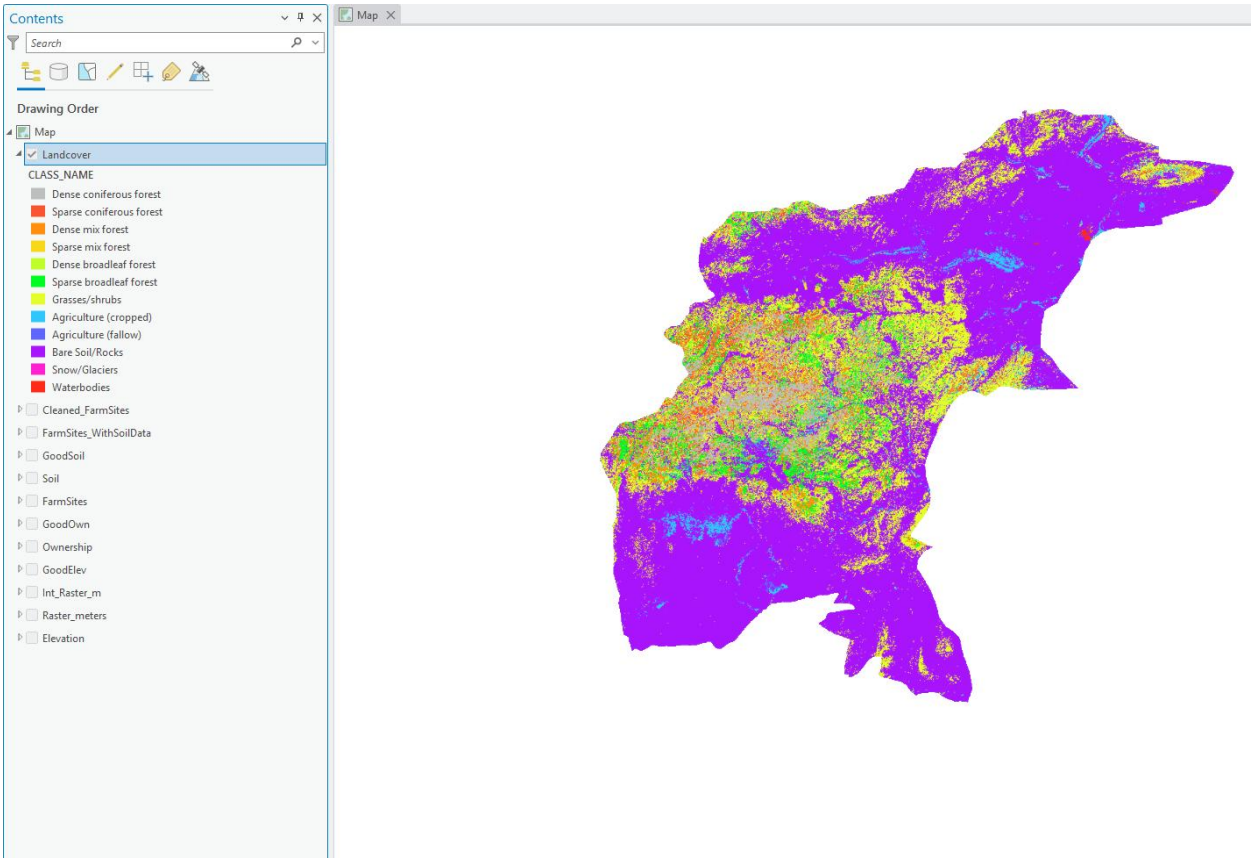
```
OutRas = Over(InRas1, InRas2)
```

Goal: Use **Over Tool**.

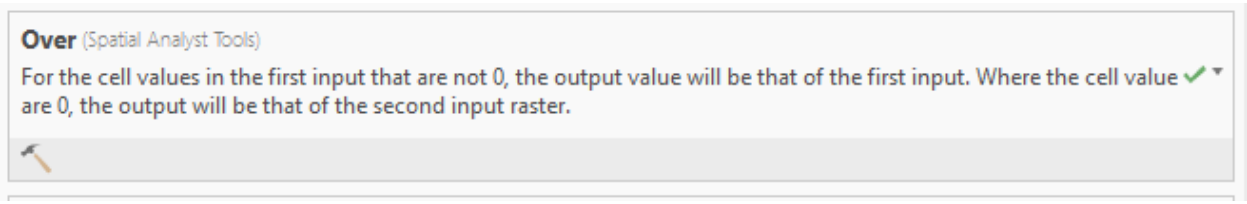
Scenario: We have an existing Land use raster, update it with the Farm Sites Data.

Load Land Use data:

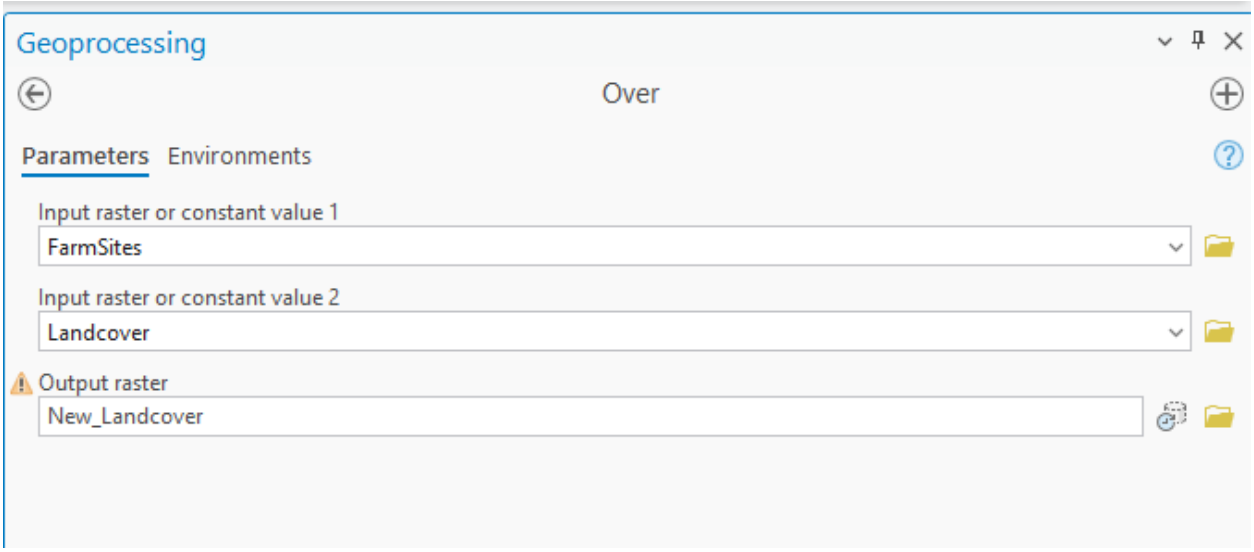




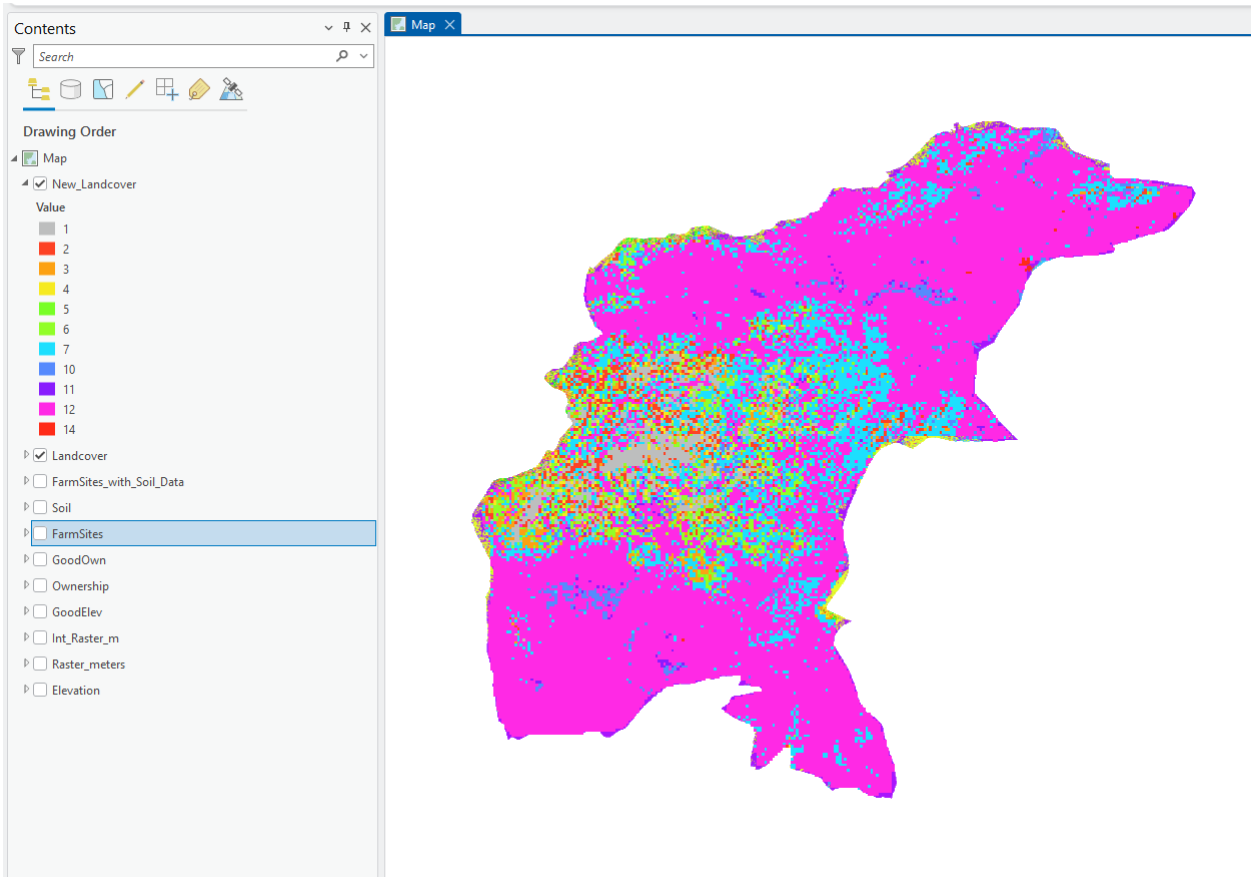
Open **Over** tool:



Configure as follows:



Hit Run:



Compare the symbology of your output layer with that of the input Landcover layer. The OVER operator has replaced the Landcover cells that coincide with non-zero cells of GoodFarm with a cell value of 1.

Diff Tool

The **Diff** tool performs a cell-by-cell logical comparison to identify differences between two input rasters.

If a cell's value in the first raster is different from the corresponding cell in the second raster, the tool outputs the value from the **first** raster.

However, if the values are identical, the output cell is assigned a **0**.

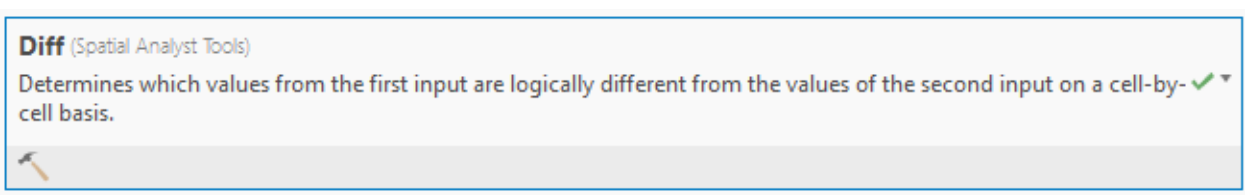
This makes it an excellent tool for change detection analysis, such as comparing land-use maps from two different years, allowing you to easily isolate areas that have changed while zeroing out regions that remained exactly the same.

```
OutRas = Diff(InRas1, InRas2)
```

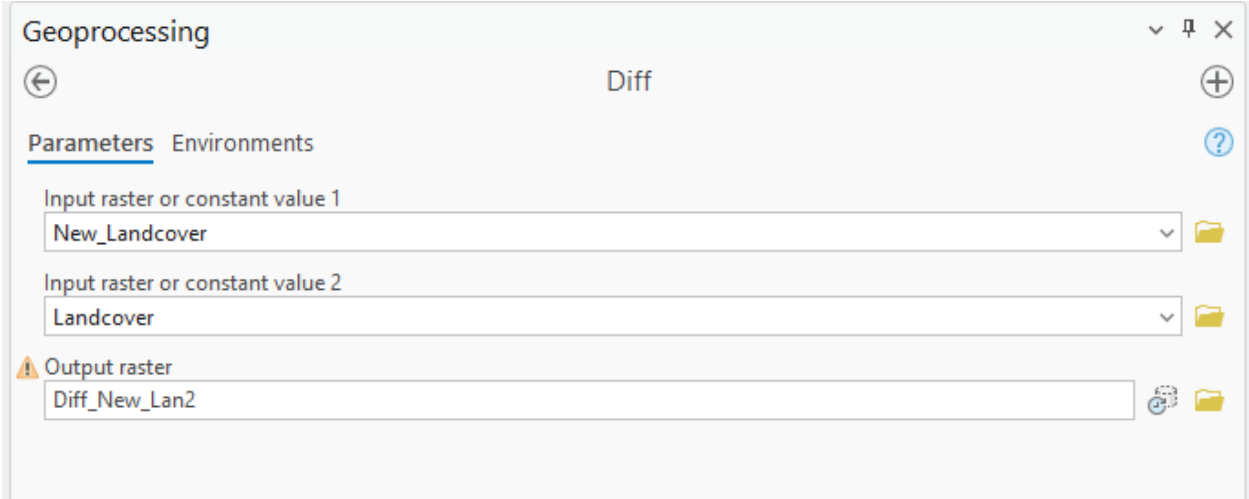
Goal: Use **Over Tool**.

Scenario: We have an existing Land use raster, update it with the Farm Sites Data.

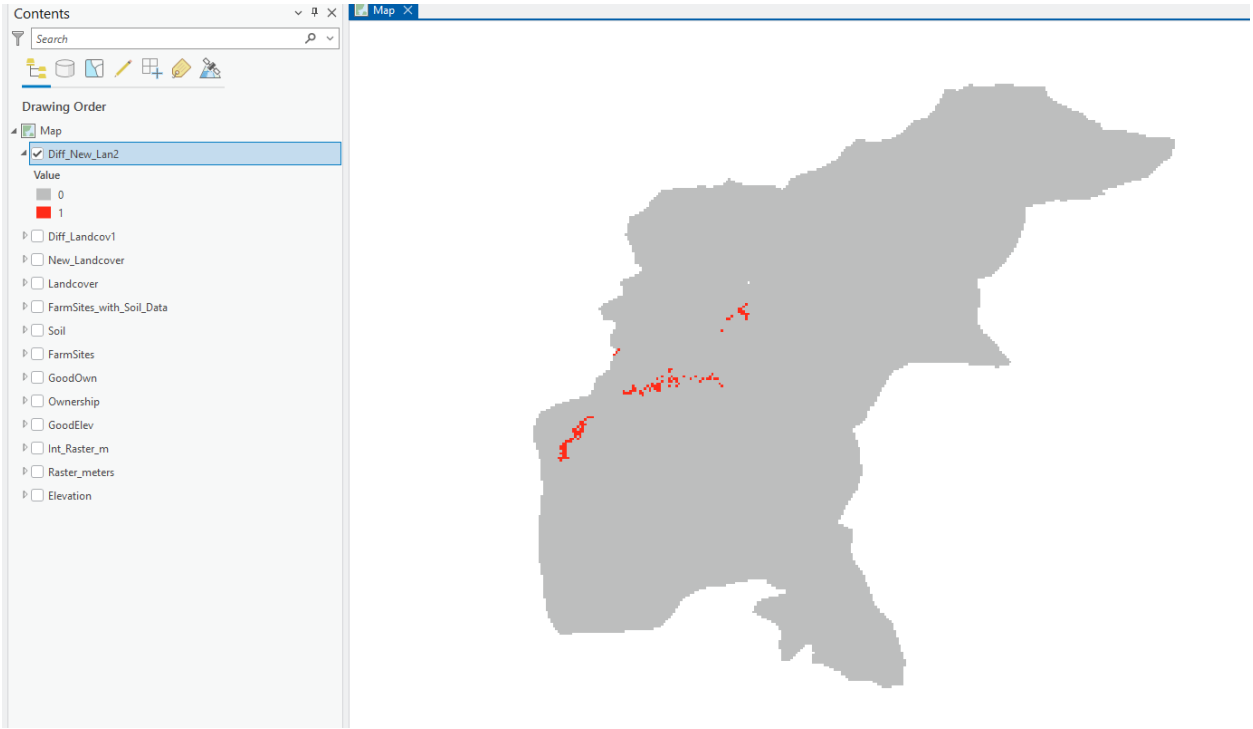
Open **Diff tool:**



Configure as follows:



Hit **Run:**



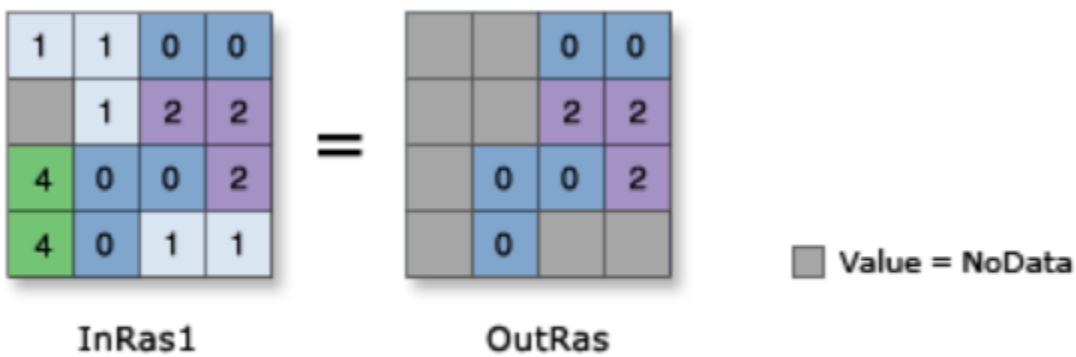
The result shows the values that were updated.

InList Tool

The **InList** tool determines which cells in an input raster match any of the values provided in a specific list of constants or other rasters.

If a cell value is found within the specified list, the output cell is assigned **1 (True)**; otherwise, it is assigned **NoData (Null)**.

It is a highly efficient way to perform **multiple "Equal To" operations at once**, such as selecting several specific land-cover classes for a focused analysis.

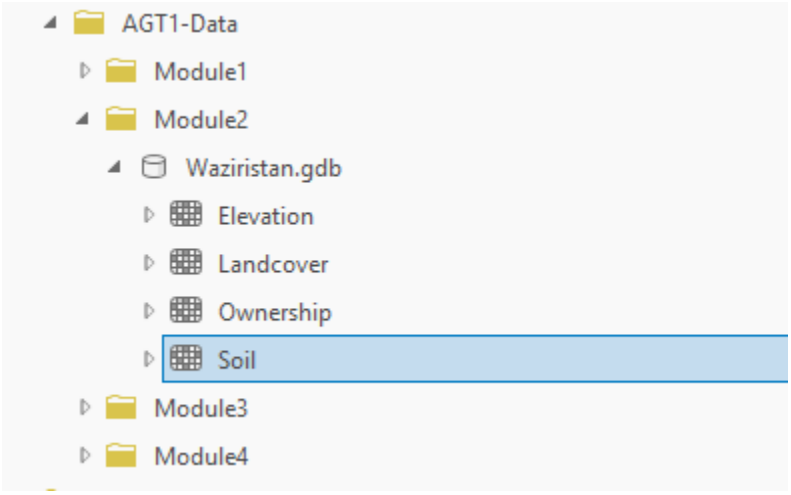


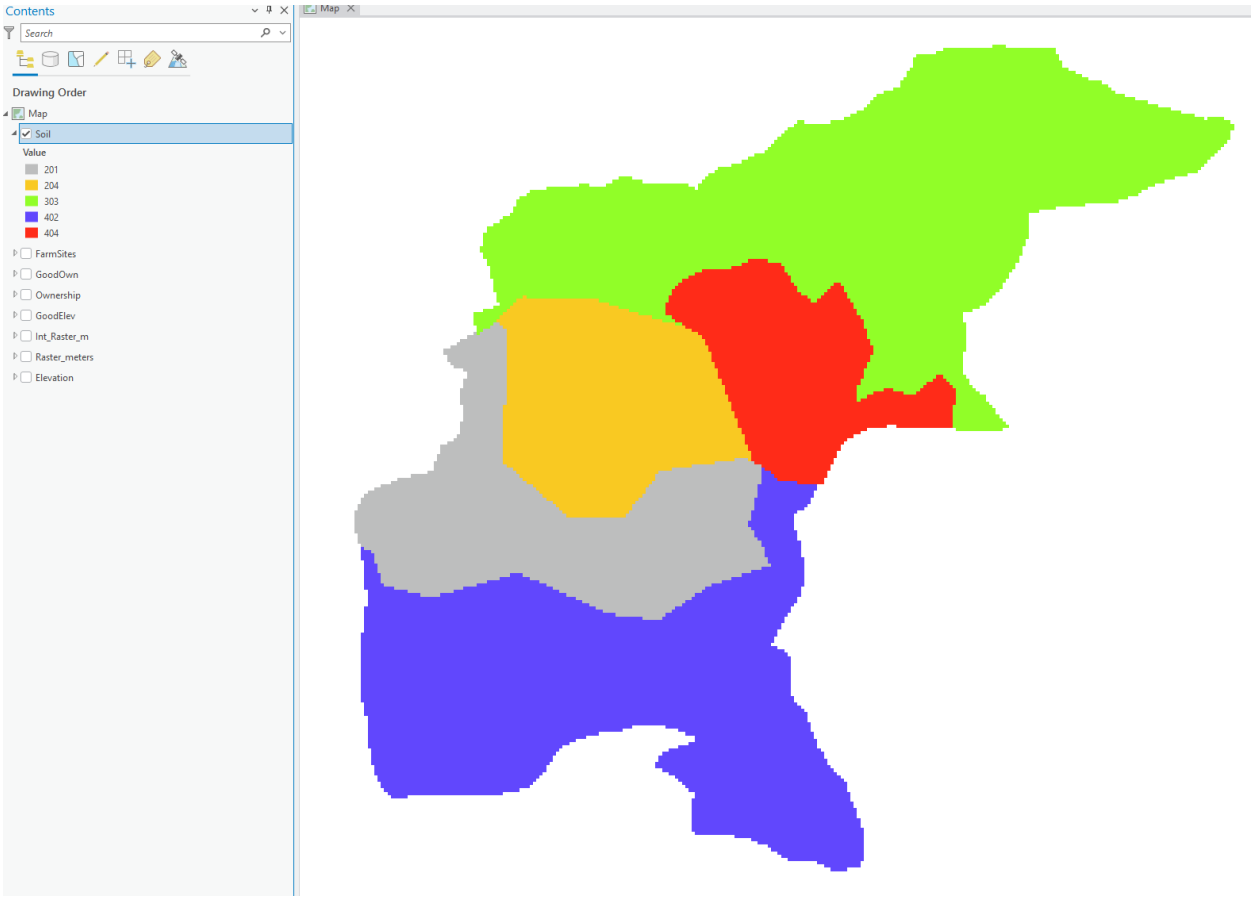

```
OutRas = InList(InRas1,[0, 2, 3, 7])
```

Goal: Use **InList Tool**.

Scenario: Certain soils, like Loam or Loamy Sand, are preferred for these species. Identify areas with only such soil types.

Load **Soils (Raster):**





Explore attribute table:

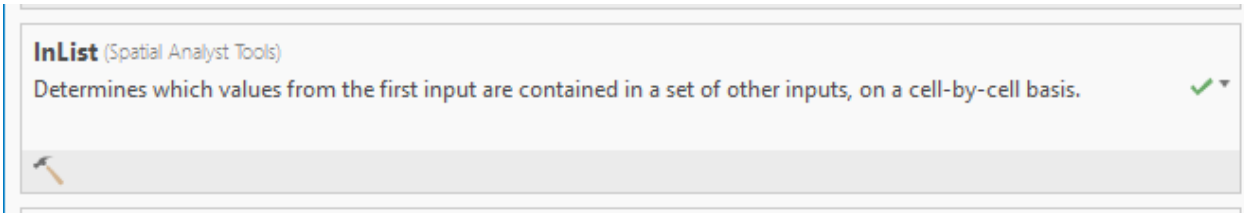
Soil

Field: Add Calculate Sel

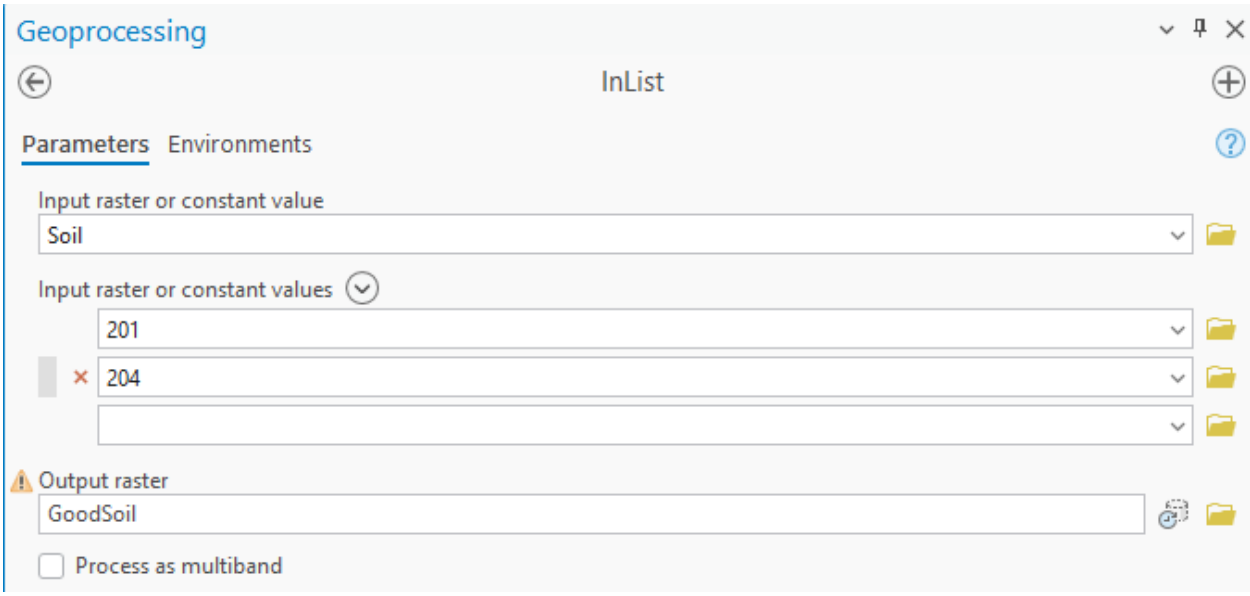
	OBJECTID *	Value	Count
1	1	201	4628
2	2	204	3638
3	3	303	10604
4	4	402	8710
5	5	404	2939
	Click to add new row.		

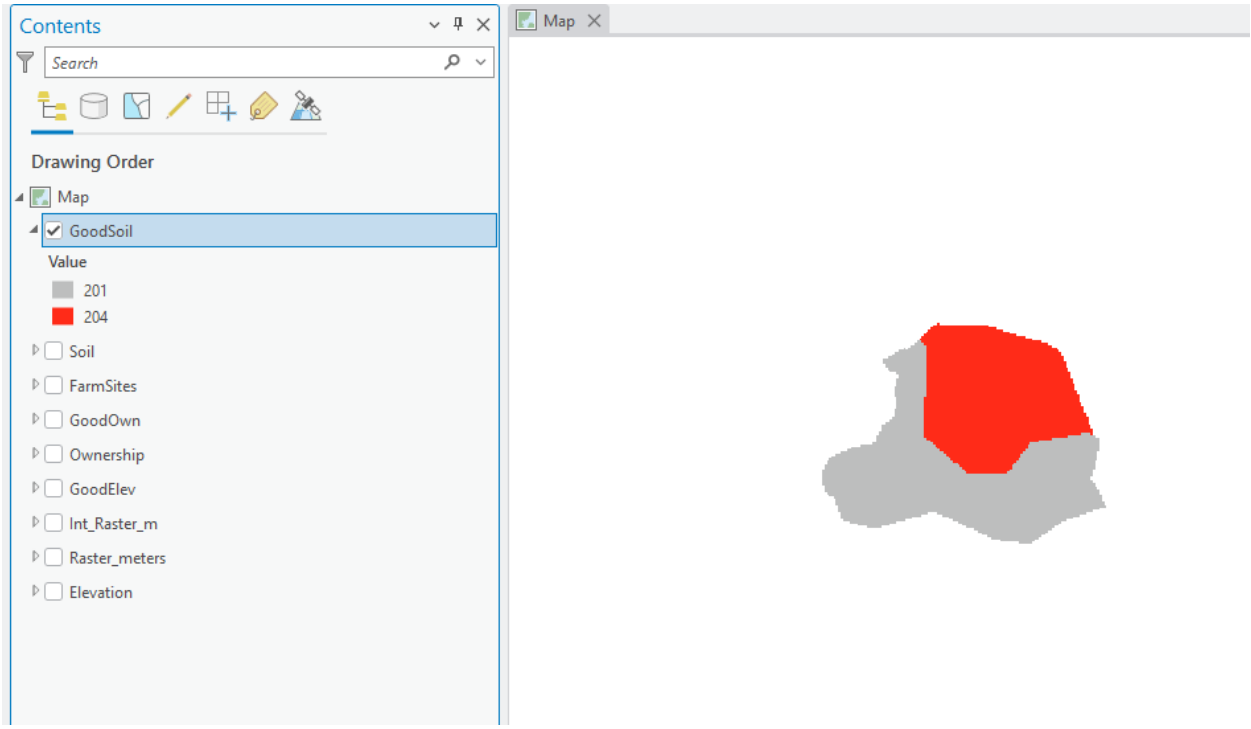
Note that in this exercise 201 & 204 correspond to Loam & Loamy sand. (The desired soils).

So we use the InList tool to find only these two soils:



Configure as follows:





Note that it assigned **NoData** to all other cells.

Deliverables (To be submitted)

- Map showing use of each tool/operator. (Screenshot **not** Map Layout)